Data Profiling Report

Apex Manufacturing Executive Dashboard

Project Name: Apex Manufacturing Executive Dashboard
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1. Purpose

The purpose of this Data Profiling Report is to evaluate the quality, structure, and integrity of the manufacturing dataset before it is used for data modeling, analysis, and dashboard development in Microsoft Power BI.

Data profiling helps identify potential data quality issues and ensures that the dataset is suitable for reliable business reporting.

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2. Dataset Overview

Attribute	Details
Industry	Manufacturing
Data Source	Microsoft Excel Workbook
Analysis Tool	Microsoft Power BI
Number of Tables	10
Data Model	Star Schema
Dashboard Pages	6

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3. Tables Included

The dataset contains the following business entities:

Table Name	Category
Plants	Dimension
Products	Dimension

Table Name	Category
Employees	Dimension
Machines	Dimension
Suppliers	Dimension
Production	Fact
Quality	Fact
Inventory	Fact
Purchase Orders	Fact
Maintenance	Fact

4. Profiling Objectives

The following data quality checks were performed:

- Verify table structure
- Validate column data types
- Review primary and foreign keys
- Check for missing values
- Identify duplicate records
- Assess data consistency
- Validate date fields
- Review numeric measures
- Verify relationship readiness
- Confirm dashboard suitability

5. Data Type Validation

The dataset contains standard analytical data types.

Data Type	Business Usage
Integer	IDs, quantities, targets
Decimal	Costs, prices, efficiency metrics
Date	Production, maintenance, procurement, inspection dates
Text	Names, categories, locations, departments

All columns were reviewed to ensure appropriate data types before loading into the Power BI data model.

6. Key Validation Checks

Primary Keys

Each dimension and fact table contains a unique identifier used for establishing relationships within the data model.

Examples include:

- Plant_ID
- Product_ID
- Employee_ID
- Machine_ID
- Supplier_ID
- Production_ID
- Inspection_ID
- Inventory_ID
- PO_ID
- Maintenance_ID

Foreign Keys

Foreign keys were verified to support one-to-many relationships between dimension and fact tables.

Examples include:

- Production → Plants
- Production → Products
- Production → Machines
- Inventory → Products
- Purchase Orders → Suppliers
- Maintenance → Machines
- Quality → Products

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7. Data Quality Assessment

The following aspects were evaluated during profiling:

Assessment Area	Status
Table Structure	Reviewed
Column Names	Standardized
Data Types	Validated

Assessment Area	Status
Key Relationships	Verified
Date Fields	Reviewed
Numeric Measures	Validated
Business Consistency	Reviewed

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8. Business Rule Validation

The dataset was reviewed to ensure that business rules are logically represented.

Examples include:

- Production quantities are associated with valid plants and products.
- Maintenance activities reference existing machines.
- Purchase orders reference valid suppliers.
- Inventory records are linked to products.
- Quality inspections correspond to manufactured products.

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9. Readiness for Data Modeling

The profiling process confirmed that the dataset is suitable for Power BI implementation and supports:

- Star schema modeling
- Relationship creation
- KPI calculations
- DAX measures
- Interactive filtering
- Business reporting

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10. Profiling Summary

The data profiling process confirmed that the manufacturing dataset has a well-defined structure suitable for analytical reporting. Tables are organized into logical business entities, data types are appropriate for analysis, and relationships support efficient Power BI modeling.

The dataset provides a solid foundation for developing interactive dashboards and business intelligence solutions while maintaining consistency across production, quality, maintenance, inventory, and procurement domains.

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11. Recommendations

To maintain long-term data quality, the following practices are recommended:

- Validate new data before importing.
- Apply consistent naming conventions.
- Enforce primary key uniqueness.
- Monitor relationship integrity.
- Standardize date formats.
- Periodically review business rules.
- Perform routine data quality checks before dashboard refreshes.

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12. Conclusion

The Apex Manufacturing dataset has been profiled to evaluate its quality, consistency, and analytical readiness. The profiling process confirms that the dataset is appropriately structured for Microsoft Power BI and supports the creation of reliable, interactive, and business-focused dashboards.

This profiling activity forms an essential step in the overall data analytics lifecycle and contributes to the accuracy and credibility of the final business intelligence solution.